

# System Analysis in Z domain

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SDU Robotics

# Digital Signal Processing Systems, Basic Filtering Types, and Digital Filter Realizations

## Reference book

Chapter 6:

6.1 The difference equation and digital filtering

6.2 Difference equation and transfer function

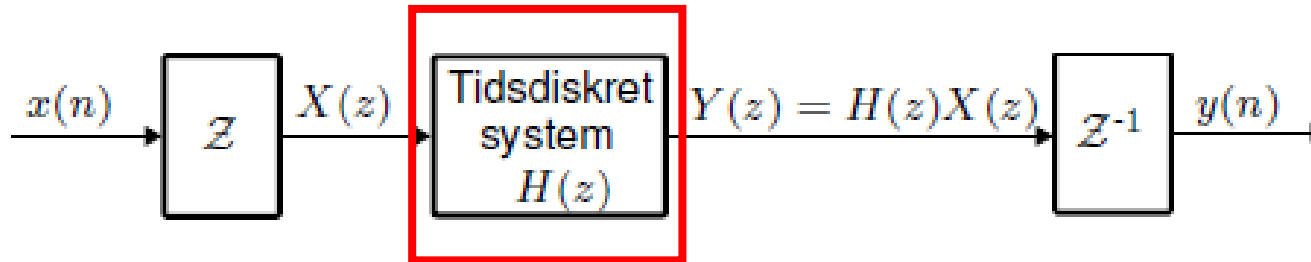
6.3 the z-plane pole-zero plot and stability

6.4 digital filter frequency response

(Page 161-186)

# Z domain in discrete signal processing

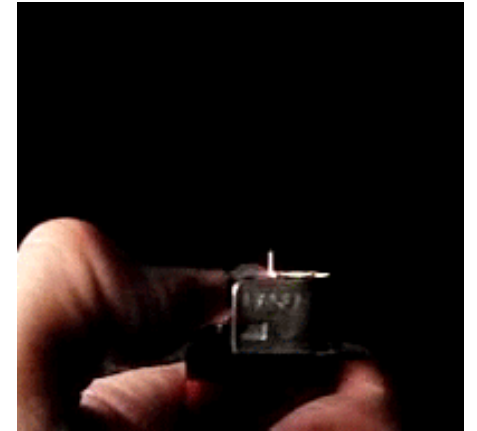
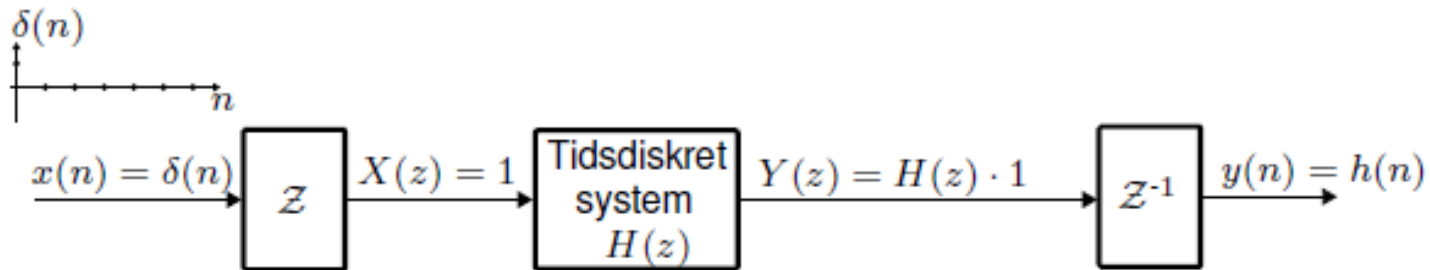
- Perform z transform on input signal  $x(n)$
- Describe the discrete time system in z domain  $H(z)$
- Obtain output  $Y(z) = H(z)X(z)$
- Perform inverse z transform on  $Y(z)$  to obtain  $y(n)$



# Impulse response

The impulse response of a time-discrete system, denoted as  $h(n)$ , is the system's output when the input is an impulse  $\delta(n)$ . It provides a fundamental description of the system.

The principle is illustrated in the following figure.



# Impulse response

Given that  $Z(\delta(n)) = 1$ , the output response in z-domain is

$$Y(z) = H(z)X(z) = H(z)$$

The impulse response sequence can be calculated by inverse z-transform of  $H(z)$ , i.e.

$$h(n) = Z^{-1}(H(z))$$

A system's impulse response sequence  $h(n)$  is found by inverse z-transform of the system's transfer function  $H(z)$ .

# Example

Considering the following transfer function, and determine the impulse response sequence of the system  $h(n)$ .

$$H(z) = \frac{1 + 0.4z^{-1}}{1 - 0.7z^{-1} + 0.1z^{-2}}$$

Firstly, we need to rewrite the transfer function with position powers

$$H(z) = \frac{z^2 + 0.4z}{z^2 - 0.7z + 0.1}$$

The transfer function can be factorized as

$$H(z) = \frac{z(z + 0.4)}{(z - 0.5)(z - 0.2)}$$

It can be seen that it has two zeros  $z_1 = 0$  and  $z_2 = -0.4$

And two poles  $p_1 = 0.5$  and  $p_2 = 0.2$ .

# Example

Through partial fraction solution

$$\frac{H(z)}{z} = \frac{k_1}{z - p_1} + \frac{k_2}{z - p_2}$$

Where

$$k_1 = (z - p_1) \frac{H(z)}{z} \Big|_{z=p_1} = \frac{z + 0,4}{z - 0,2} \Big|_{z=0,5} = 3$$

$$k_2 = (z - p_2) \frac{H(z)}{z} \Big|_{z=p_2} = \frac{z + 0,4}{z - 0,5} \Big|_{z=0,2} = -2$$

Thus, the transfer function becomes

$$H(z) = 3 \frac{z}{z - 0,5} - 2 \frac{z}{z - 0,2}$$

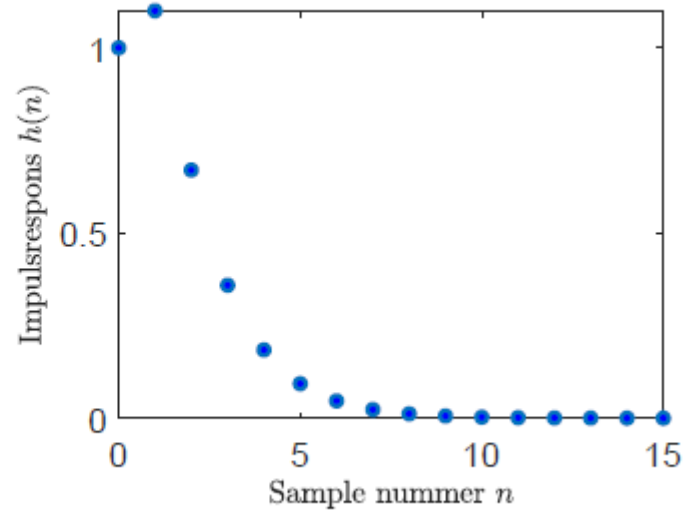
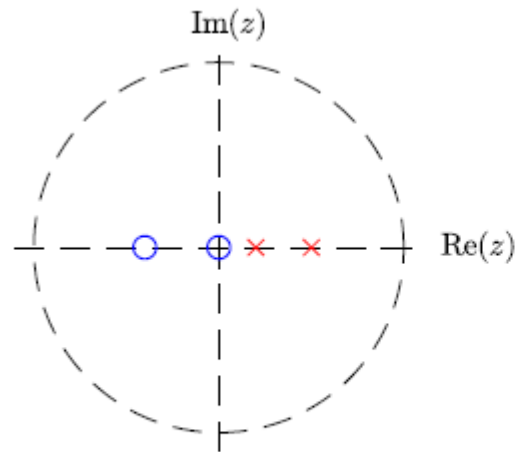
Inverse z-transform

$$h(n) = \mathcal{Z}^{-1}[H(z)] = 3 \cdot 0,5^n - 2 \cdot 0,2^n = 3e^{-0,693n} - 2e^{-1,61n}$$

# We can draw it

$$H(z) = \frac{z(z + 0,4)}{(z - 0,5)(z - 0,2)}$$

$$h(n) = \mathcal{Z}^{-1}[H(z)] = 3 \cdot 0,5^n - 2 \cdot 0,2^n = 3e^{-0,693n} - 2e^{-1,61n}$$



# Matlab function: tf()

'tf()' is used to create a transfer function, either for **continuous-time system**, or **discrete-time system**.

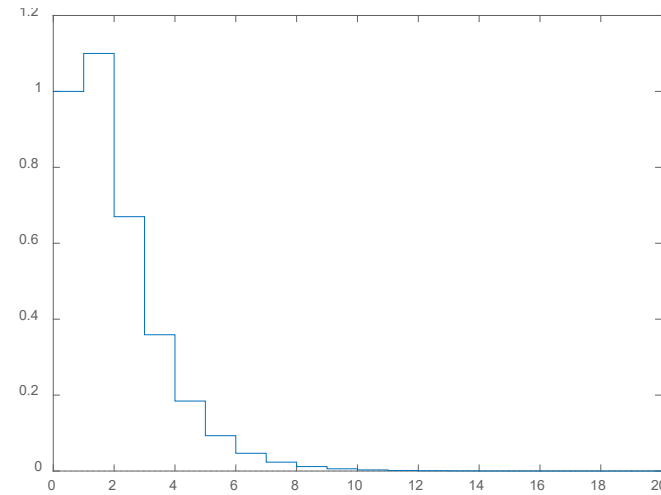
→ **continuous-time system**: `sys = tf(numerator,denominator)`

→ **discrete-time system**: `sys = tf(numerator,denominator, ts)`

To leave the sample time unspecified, set 'ts' to -1.

# Matlab function: impulse()

```
H_z = tf([1, 0.4, 0],[1, -0.7, 0.1], -1)  
n=0:20;  
impulse(H_z,n)
```



# Let's take a closer look

A transfer function can be factorized as

$$H(z) = a_0 \frac{(z - z_1)(z - z_2) \dots (z - z_N)}{(z - p_1)(z - p_2) \dots (z - p_N)}$$

where  $z_i$  for  $i = 1, \dots, N$  are zeros, and  $p_i$  for  $i = 1, \dots, N$  are poles

Using partial fraction splitting, the transfer function can be written

$$\frac{H(z)}{z} = \frac{k_1}{z - p_1} + \frac{k_2}{z - p_2} + \dots + \frac{k_N}{z - p_N}$$

where  $k_1, k_2, \dots, k_N$  are coefficients

Thus, the impulse response sequence can be written as

$$h(n) = h_1(n) + h_2(n) + \dots + h_N(n)$$

Given the pole  $p_i$  the following relationship:

$$z = p_i = e^{s_i T} = e^{\sigma_i T} e^{j\omega_i T}$$

We have

$$h_i(n) = \mathcal{Z}^{-1} \left[ \frac{z}{z - p_i} \right] = e^{s_i n T} = e^{\sigma_i n T} e^{j\omega_i n T}$$

Assuming it has 'Simple poles', meaning that all poles have multiplicity of 1.

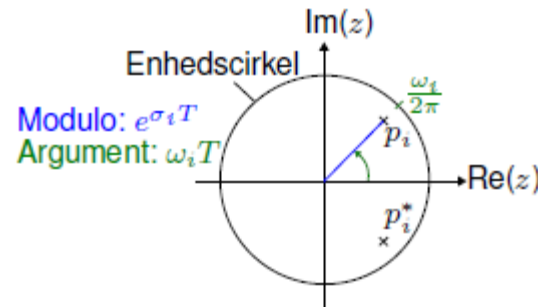
# Relation between pole placement and impulse response

Let's consider the following impulse response

$$h_i(n) = e^{\sigma_i nT} e^{j\omega_i nT} = e^{\sigma_i nT} \angle \omega_i nT$$

|                                                    |
|----------------------------------------------------|
| Euler's formula:<br>$e^{ix} = \cos(x) + i \sin(x)$ |
|----------------------------------------------------|

Therefore, the pole  $p_i$  can be written in polar form as  $p_i = e^{\sigma_i T} \angle \omega_i T$



From the above impulse response sequence  $h_i(n)$  :

- **Modulus** of  $h_i(n)$  changes by a factor  $e^{\sigma_i T}$  **between two consecutive samples.**
- **Argument** of  $h_i(n)$  is changed by  $\omega_i T$  **between two consecutive samples.**

# Example: Impulse response of a 2<sup>nd</sup> order system

Let's consider the following 2<sup>nd</sup> order system

$$H(z) = \frac{z}{z^2 + 1.697z + 1.44}$$

The transfer function  $H(z)$  has 1 zero  $z = 0$ , and 2 poles  $p_1 = 1.2 \cdot e^{j3\pi/4}$  and  $p_2 = 1.2 \cdot e^{-j3\pi/4}$

The impulse response for the system is calculated via partial fraction

$$\frac{H(z)}{z} = \frac{k_1}{z - p_1} + \frac{k_2}{z - p_2}$$

where

$$k_1 = (z - p_1) \frac{H(z)}{z} \Big|_{z=p_1} = \frac{1}{p_1 - p_2} = \frac{1}{1.2(e^{j3\pi/4} - e^{-j3\pi/4})} = \frac{1}{2.4j \sin\left(\frac{3\pi}{4}\right)} = \frac{1}{2.4 \sin\left(\frac{3\pi}{4}\right)} e^{-j\pi/2}$$

$k_1$  and  $k_2$  are conjugate pair

Euler's formula:

- $e^{ix} = \cos(x) + i \sin(x)$
- $j = e^{j\pi/2}$

# Example: Impulse response of a 2<sup>nd</sup> order system

The impulse response can be calculated via inverse z-transformation

$$\begin{aligned} h(n) &= \frac{1}{2.4 \sin\left(\frac{3\pi}{4}\right)} \left( e^{-\frac{j\pi}{2}} 1.2^n e^{jn\frac{3\pi}{4}} + e^{\frac{j\pi}{2}} 1.2^n e^{-jn\frac{3\pi}{4}} \right) \\ &= \frac{1}{2.4 \sin\left(\frac{3\pi}{4}\right)} \mathbf{1.2^n} \left( e^{j\left(n\frac{3\pi}{4} - \frac{\pi}{2}\right)} + e^{-j\left(n\frac{3\pi}{4} - \frac{\pi}{2}\right)} \right) \end{aligned}$$

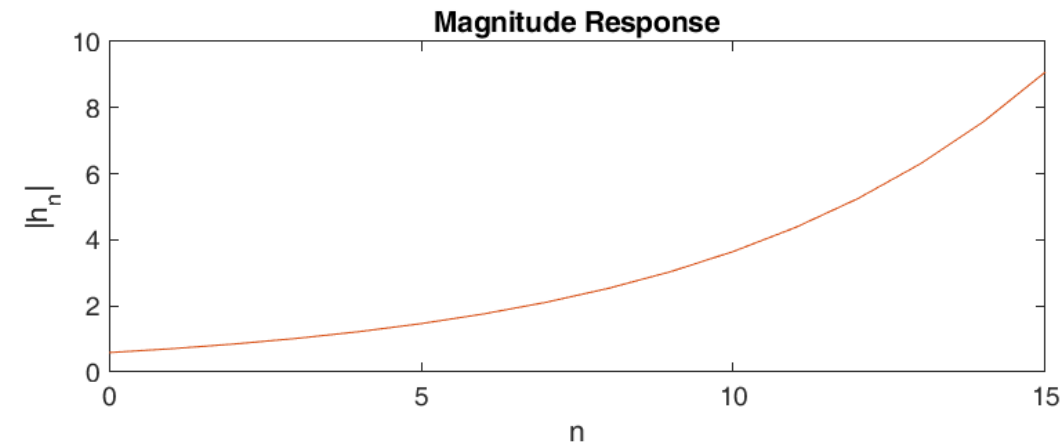
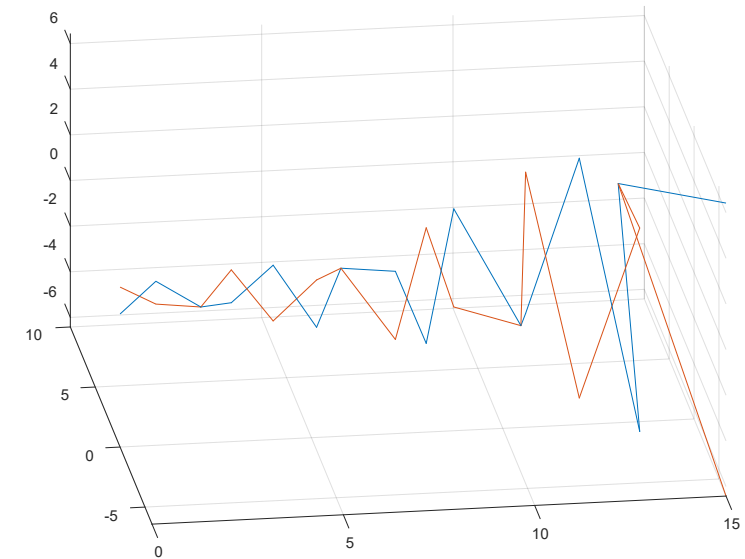
| $x(n)$ | $X(z)$            |
|--------|-------------------|
| $a^n$  | $\frac{z}{z - a}$ |

# Example: Impulse response of a 2<sup>nd</sup> order system

```
% plot it out
n = 0:1:15;
h1_n = 1/(2.4*sin(3*pi/4)) * 1.2.^n .* exp(1j*(3*pi/4*n - pi/2));
h2_n = 1/(2.4*sin(3*pi/4)) * 1.2.^n .* exp(-1j*(3*pi/4*n - pi/2));
figure()
plot3(n,real(h1_n),imag(h1_n))
hold on
plot3(n,real(h2_n),imag(h2_n))
grid on
```

```
n = 0:15;
H_n = h1_n + h2_n;
plot(h)
```

The output  $h(n)$  keeps increasing with  $n$ .  
**This system is not stable!**



# Stability

A system's stability state can be one of the following

→ **Stable system**: A system is stable if its impulse response  $h(n)$  goes to zero as  $n$  goes to infinity

$$|h(n)| \rightarrow 0 \quad \text{for } n \rightarrow \infty$$

→ **Marginally stable system**: A system is marginally stable if its impulse response  $h(n)$  goes towards constant value different from zero or oscillates with constant amplitude and frequency as  $n$  goes towards infinity.

→ **Unstable system**: A system is unstable if its impulse response  $h(n)$  grows without limit as  $n$  goes to infinity

$$|h(n)| \rightarrow \infty \quad \text{for } n \rightarrow \infty$$

# Determination of stability

Transfer function  $H(z)$  has poles  $p_1, p_2, \dots, p_N$ .

→ The system is **stable** if all poles lie within the unit circle, i.e.

$$|p_i| < 1 \text{ for } i = 1, 2, \dots, N$$

→ The system is **marginally stable** if at least one pole (e.g.  $p_j$ ) lies on the unit circle, while the other poles lie within the unit circle, i.e.

$$|p_i| \leq 1 \text{ for } i = 1, 2, \dots, N$$

or

$$|p_j| = 1 \text{ for } j \in \{1, 2, \dots, N\}$$

→ The system is unstable if a pole (e.g.  $p_j$ ) lies outside the unit circle, i.e.

$$|p_j| > 1 \text{ for } j \in \{1, 2, \dots, N\}$$

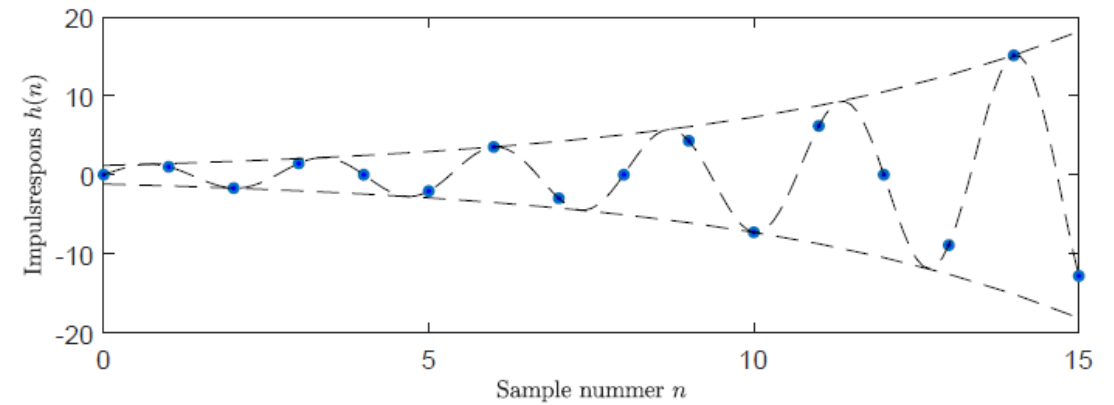
# The previous example

The transfer function is  $H(z) = \frac{z}{z^2 + 1.697z + 1.44}$

Its poles are

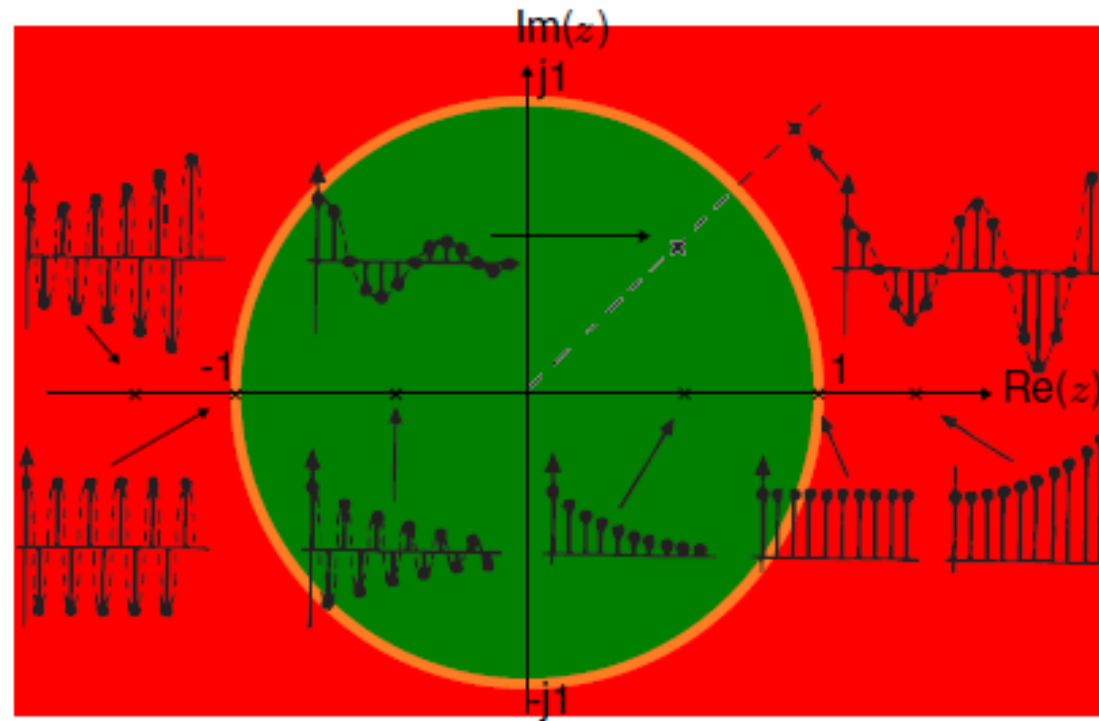
$$p_1 = 1.2 \cdot e^{j3\pi/4} \text{ and } p_2 = 1.2 \cdot e^{-j3\pi/4}$$

And  $|p| > 1$  applies!

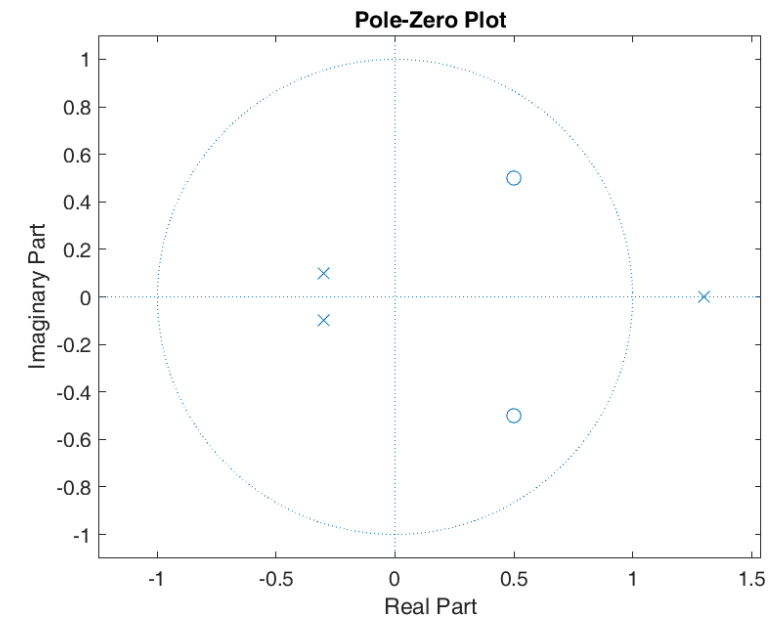
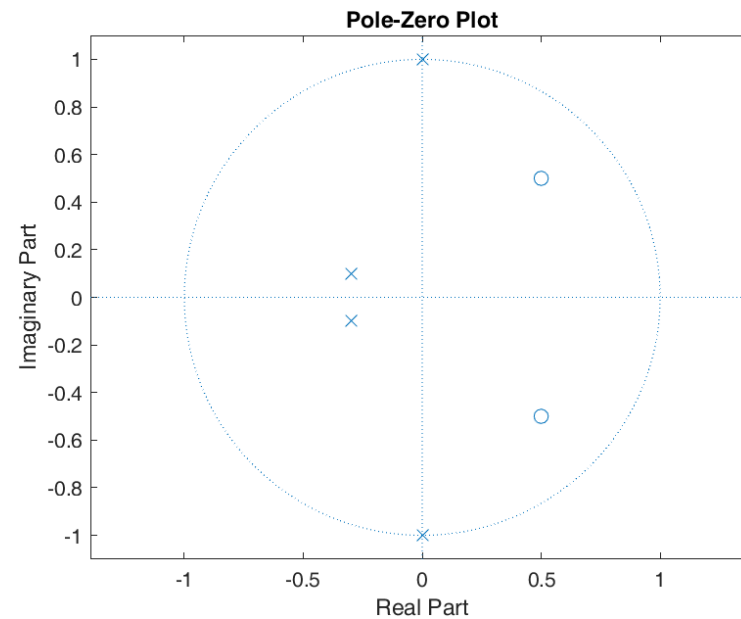
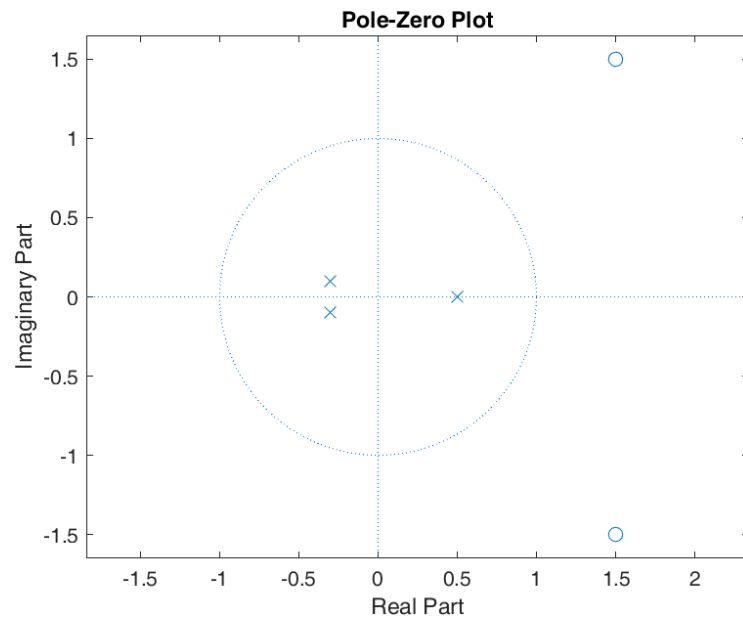


# Impulse response in z-plane related to pole position

$$\text{pole } p_i = e^{\sigma_i T} e^{j\omega_i T}$$

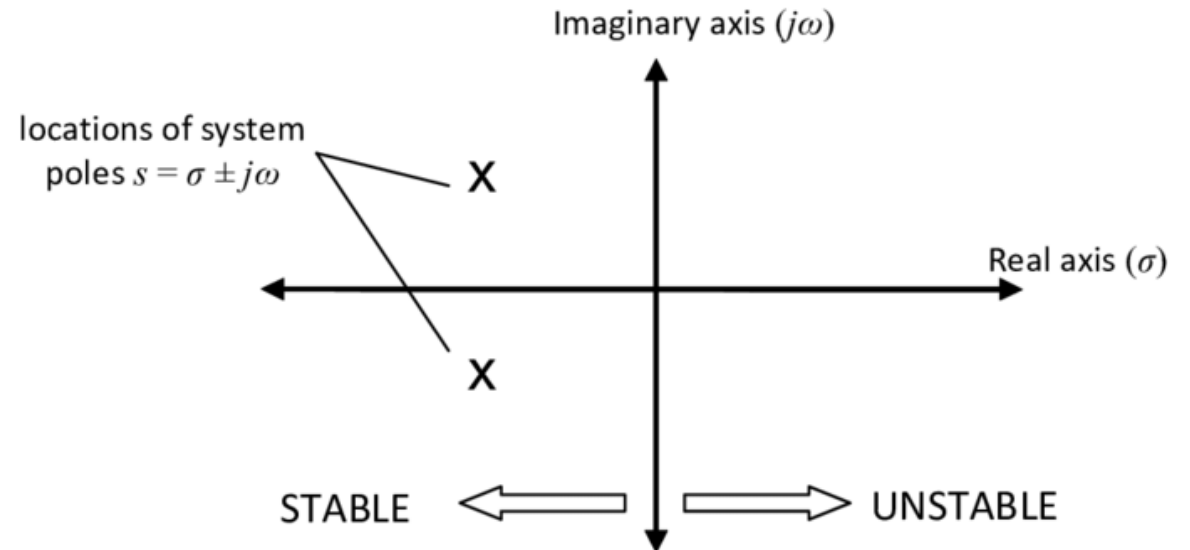
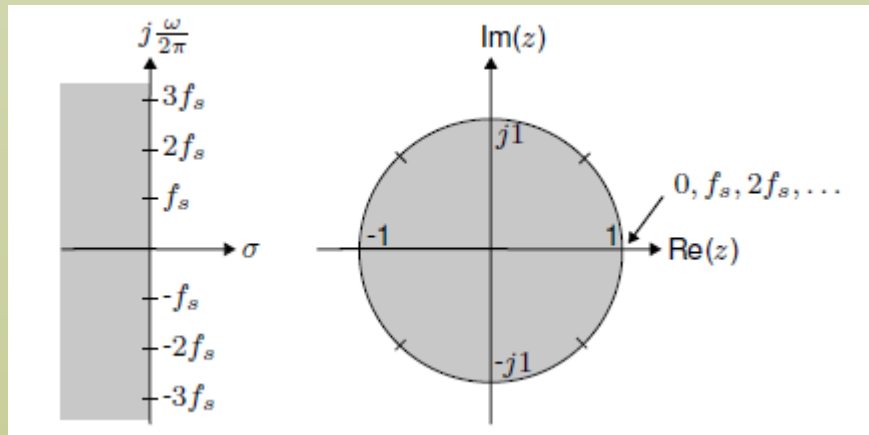


# Are these systems stable?

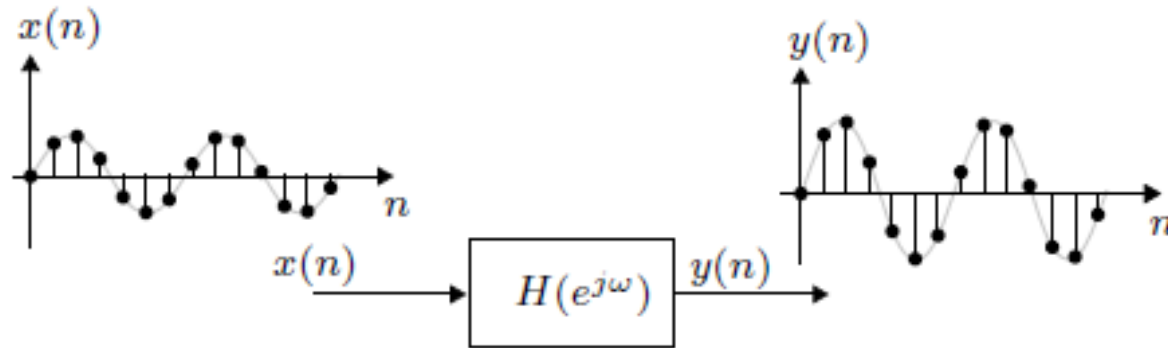


# What about in continuous time system (s-domain)?

Given the following projection relationship:



# Frequency response analysis



A frequency response analysis provides the response of a system to a sinusoidal input sequence. Here it is assumed that the sinusoidal sequence has been imprinted from time  $-\infty$  (the analysis disregards transient response).

# Frequency response analysis

A system's frequency response is the response (output signal) when a **sinusoidal input** is applied to a system.

The output signal of a system with **input signal**  $A \cos(\omega t)$  can be determined as

$$y(t) = H(\cdot) x(t) = H(\cdot) A \cos(\omega t)$$

Given Euler equation:

$$A \cos(\omega t) = \frac{A}{2} (e^{j\omega t} + e^{-j\omega t})$$

The response can be written as

$$y(t) = \frac{A}{2} (H(j\omega)e^{j\omega t} + H(-j\omega)e^{-j\omega t})$$

# Frequency response analysis

$$y(t) = \frac{A}{2} (H(j\omega)e^{j\omega t} + H(-j\omega)e^{-j\omega t})$$

In polar form,  $H(j\omega) = M(\omega)e^{j\varphi(\omega)}$ , so

$$y(t) = \frac{A}{2} M(\omega) (e^{j(\omega t + \varphi(\omega))} + e^{-j(\omega t + \varphi(\omega))}) = AM(\omega)\cos(\omega t + \varphi(\omega))$$

where

$$M(\omega) = |H(j\omega)| \quad \text{and} \quad \varphi(\omega) = \angle H(j\omega)$$

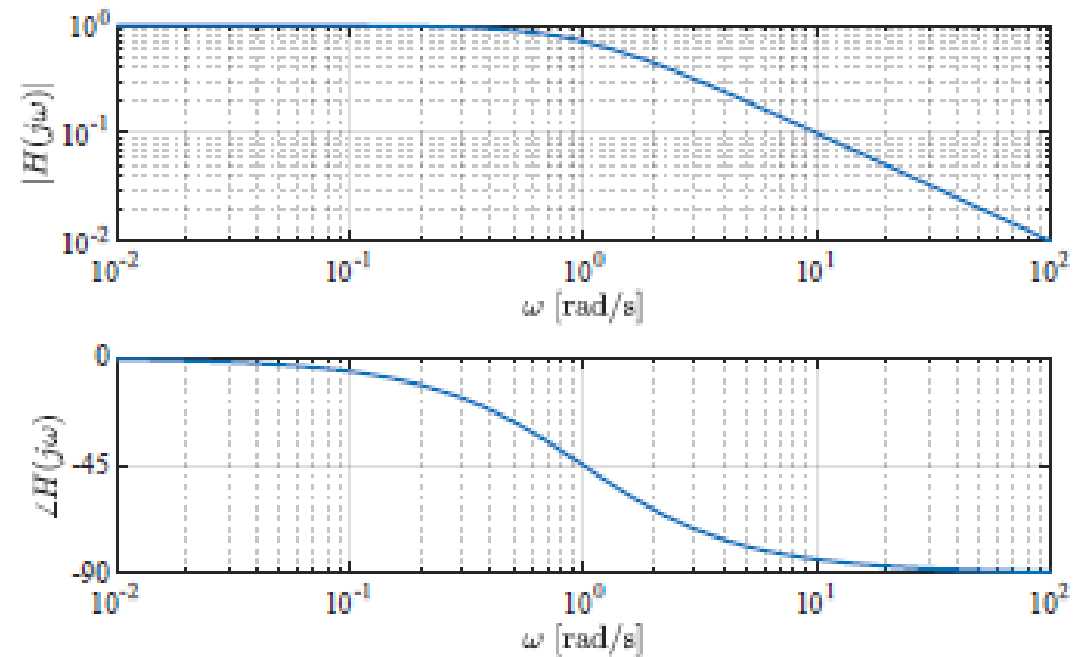
# Bode plot

A **Bode plot** is used to visualize the frequency response, and is usually drawn on a logarithmic scale.

$$y(t) = AM(\omega)\cos(\omega t + \varphi(\omega))$$

where

$$\mathbf{M}(\omega) = |H(j\omega)| \quad \text{and} \quad \mathbf{\varphi}(\omega) = \angle H(j\omega)$$



# Hand calculation of frequency response

To study the frequency response of a system, we only look at values of  $z$  that lie on the unit circle, i.e.

$$z = e^{j\omega T}$$

The frequency response of a time-discrete system  $H(z)$  is thus given as

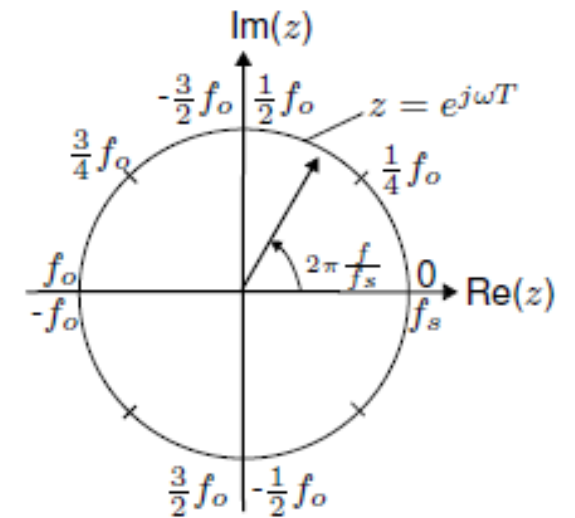
$$H(z) \Big|_{z = e^{j\omega T}} = H(e^{j\omega T}) = \frac{Y(e^{j\omega T})}{X(e^{j\omega T})}$$

In polar form, the frequency response is

$$H(j\omega) = |H(\omega)| \angle \varphi(\omega)$$

The **amplitude** is normally given in dB, i.e.

$$|H(\omega)| = 20 \log \frac{|Y(j\omega)|}{|X(j\omega)|} \quad [\text{dB}]$$



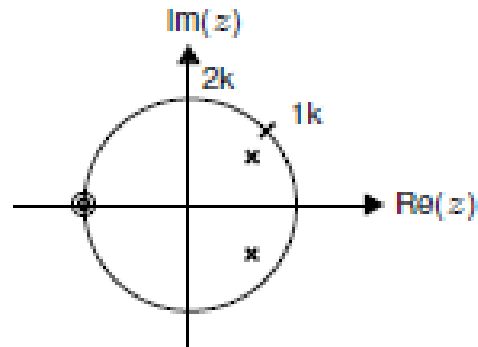
**Ultra cautious!**

The 'log()' function in Matlab means  $y = \ln(x)$ .  
You should use the 'log10()' function

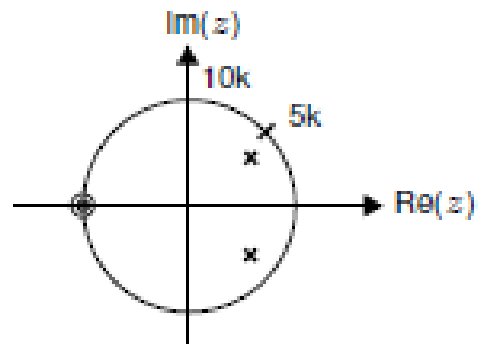
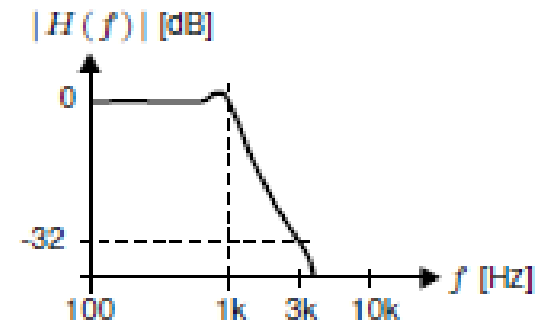
# Sample rate

The importance of the sample frequency for the frequency response is only to move the graph along the frequency axis.

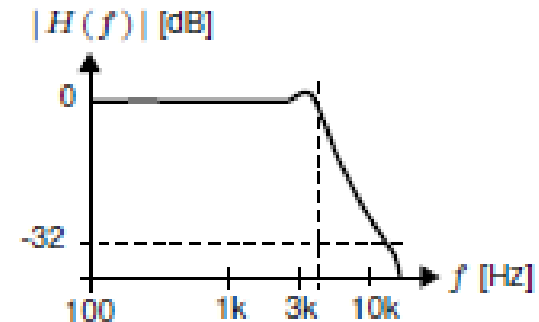
$$z = e^{j\omega T}$$



$$f_s = 8 \text{ kHz}$$



$$f_s = 40 \text{ kHz}$$



# Example

Let's consider the following transfer function

$$H(z) = \frac{z^2 + 0.4z}{z^2 - 0.7z + 0.1}$$

To determine the frequency response, we calculate  $H(j\omega)$ , i.e.

$$H(z) \Big|_{z = e^{j\omega T}} = H(e^{j\omega T}) = \frac{e^{j2\omega T} + 0.4e^{j\omega T}}{e^{j2\omega T} - 0.7e^{j\omega T} + 0.1}$$

The frequency response of input signal  $x(n) = \sin(\omega nT)$ , and  $\omega T = 1$

$$H(j\omega) = \frac{e^{j2} + 0.4e^j}{e^{j2} - 0.7e^j + 0.1} = 0.92 - j1.37 = 1.65 \angle -56^\circ$$

Use Matlab for calculation:  
`(exp(2j) + 0.4*exp(1j))/(exp(2j)-0.7*exp(1j)+0.1)`

# Example – through z transform

Let's consider the following transfer function:

$$H(z) = \frac{z^2 + 0.4z}{z^2 - 0.7z + 0.1}$$

The z-transform of the input signal  $x(n) = \sin(\omega n T)$  is

$$X(z) = \frac{\sin(\omega T) z}{z^2 - 2 \cos(\omega T) z + 1}$$

|     |                     |                                                              |
|-----|---------------------|--------------------------------------------------------------|
| ZT6 | $\sin \omega_0 n T$ | $\frac{(\sin \omega_0 T) z}{z^2 - 2(\cos \omega_0 T) z + 1}$ |
|-----|---------------------|--------------------------------------------------------------|

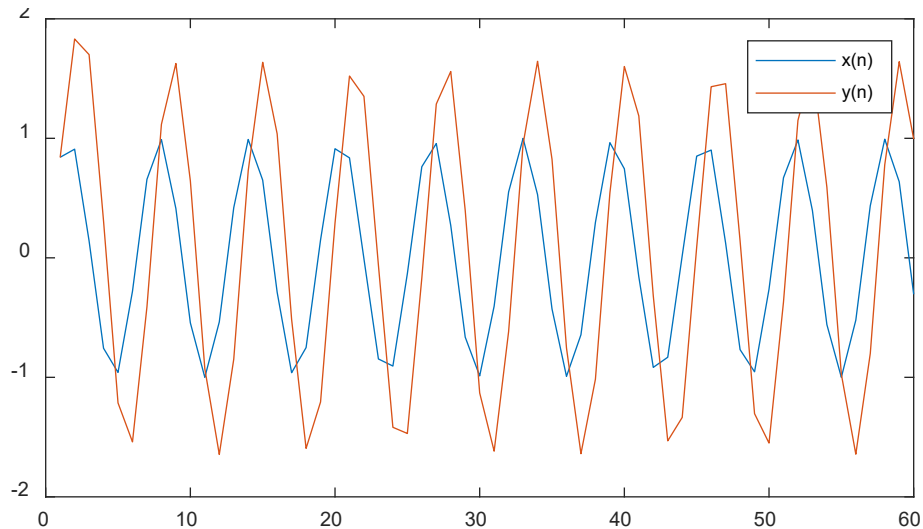
We can calculate  $Y(z)$ , and let's study an example frequency response  $\omega T = 1$

$$Y(z) = H(z)X(z) = \frac{z^2 + 0.4z}{z^2 - 0.7z + 0.1} \cdot \frac{0.84z}{z^2 - 1.08z + 1}$$

Implement inverse z-transform, and we have  $y(n)$ . Then we can plot the input and output.

# Example

Let's compare the input and output signal  $x(n) = \sin(n)$



$$H(j\omega) = 1.65\angle -56^\circ$$

Matlab function:

subs()  
lsim()

```
syms z
Y = 0.84*z*z*(z+0.4)/((z^2-
0.7*z+0.1)*(z^2-1.08*z+1));
y= iztrans(Y);

syms n
y_res = subs(y, n, {1:60});

n=1:60;
x=sin(n);

plot(n,x);
hold on
plot(n, y_res);
```

**Simulating an input signal passing through this filter:**

```
% Define time vector
t = 0:0.0001:0.01;
% Generate input signal 1000Hz
x = sin(2*pi*1000*t);
% output after passing through the filter
y = lsim(sys, x, t);

% plot both input & output
plot(t, x, 'b', t, y, 'r');
```

# Matlab function – bode()

Function 'bode()' is used to **obtain results** and **visualize** the frequency response of the system, either **continuous time system** or **discrete time system**.

## Important trick!!

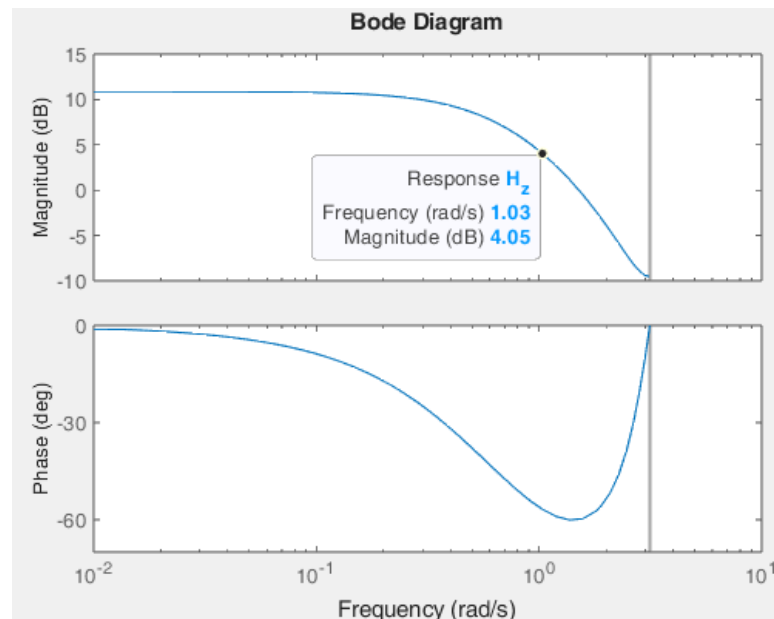
The 'bode()' function by default plots in 'rad/s'. Add the following commands to plot in 'Hz':

```
options = bodeoptions;  
options.FreqUnits = 'Hz';  
bode(H_z, options);
```

The transfer function in the example:  $H(z) = \frac{z^2 + 0.4z}{z^2 - 0.7z + 0.1}$

```
%% Matlab function  
H_z = tf([1, 0.4, 0], [1, -0.7, 0.1], -1)  
bode(H_z)
```

Be aware that we are checking  
input frequency  $\omega T = 1$



The result just now:

$$H(j\omega) = 1.65 \angle -56^\circ$$

And

$$20 \log(1.65) = 4.3497$$

# Matlab function

To see the frequency response of a filter:

`% for analog filter, magnitude (not dB)`

`freqs()`

`% for digital filter, magnitude (not dB)`

`freqz()`

# Question?

What about  $\omega T = 10$ ?

# Exercise

6.3. Given the difference equation

$$y(n) = x(n-1) - 0.75y(n-1) - 0.125y(n-2)$$

- a. calculate the system response  $y(n)$  for  $n = 0, 1, \dots, 4$  with the input  $x(n) = (0.5)^n u(n)$  and initial conditions:  $x(-1) = -1$ ,  $y(-2) = 2$ , and  $y(-1) = 1$ ;
- b. calculate the system response  $y(n)$  for  $n = 0, 1, \dots, 4$  with the input  $x(n) = (0.5)^n u(n)$  and zero initial conditions:  $x(-1) = 0$ ,  $y(-2) = 0$ , and  $y(-1) = 0$ .

6.4. Given the difference equation

$$y(n) = 0.5x(n) + 0.5x(n-1)$$

- a. find the  $H(z)$ ;
- b. determine the impulse response  $y(n)$  if the input  $x(n) = 4\delta(n)$ ;
- c. determine the step response  $y(n)$  if the input  $x(n) = 10u(n)$ .

# Exercise

**6.12.** Convert each of the following transfer functions into difference equations:

a.  $H(z) = \frac{z^2 - 0.25}{z^2 + 1.1z + 0.18}$

b.  $H(z) = \frac{z^2 - 0.1z + 0.3}{z^3}$

**6.15.** Given each of the following transfer functions that describe digital systems, sketch the z-plane pole-zero plot and determine the stability for each digital system.

a.  $H(z) = \frac{z - 0.5}{(z + 0.25)(z^2 + z + 0.8)}$

b.  $H(z) = \frac{z^2 + 0.25}{(z - 0.5)(z^2 + 4z + 7)}$

c.  $H(z) = \frac{z + 0.95}{(z + 0.2)(z^2 + 1.414z + 1)}$

d.  $H(z) = \frac{z^2 + z + 0.25}{(z - 1)(z + 1)^2(z - 0.36)}$